

$$\int_0^{256} \frac{1}{8\left(\sqrt{\sqrt{\sqrt{x}+9}-3}\right)\left(\sqrt{\sqrt{x}+9}\right)\left(\sqrt{2x}\right)}dx$$

$$\frac{1}{8\sqrt{2}}\int_0^{256}\frac{1}{\sqrt{\sqrt{\sqrt{x}+9}-3}\sqrt{\sqrt{x}+9}\sqrt{x}}dx$$

$$u:=\sqrt{\sqrt{x}+9}$$

$$du:=\frac{1}{4\sqrt{\sqrt{x}+9}\sqrt{x}}dx$$

$$\frac{1}{2\sqrt{2}}\int_3^5\frac{1}{\sqrt{u-3}}du$$

$$w:=u-3$$

$$dw:=du$$

$$\frac{1}{2\sqrt{2}}\int_0^2w^{-\frac{1}{2}}dw$$

$$\frac{w^{\frac{1}{2}}}{\sqrt{2}}\Big|_0^2$$

$$1$$